INCH-POUND

FF-B-2844/03
15 FEBRUARY 1994

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR: SINGLE ROW, RADIAL, NON-FILLING SLOT, EXTRA LIGHT SERIES (R - SERIES) (LARGER THAN 0.7500 BORE) (CHROMIUM ALLOY STEEL)

The General Services Administration has authorized the use of this specification by all Federal Agencies.

The complete requirments for procuring the bearings described herein shall consist of this document and the latest issue of FF-B-2844/GEN.

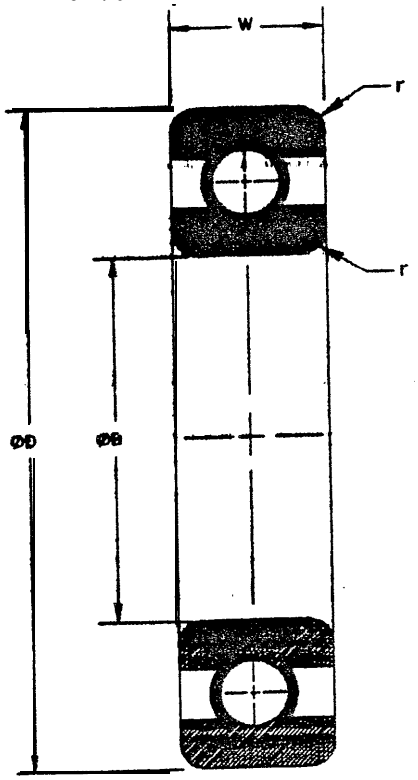


FIGURE 1.
OPEN(NO SHIELDS OR SEALS)

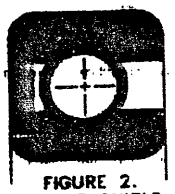


FIGURE 2.
SINGLE SHIELD

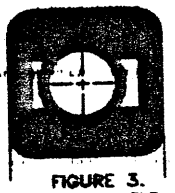


FIGURE 3.
DOUBLE SHIELD

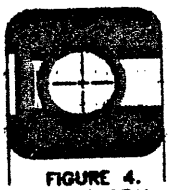


FIGURE 4.
SINGLE SEAL

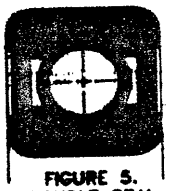


FIGURE 5.
DOUBLE SEAL



FIGURE 6.
SINGLE SHIELD AND SEAL

BEARING CONFIGURATIONS

FSC 3110

AMSC/NA

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REQUIREMENTS

1. **Boundary dimensions**. The boundary dimensions shall be specified in the part number by the bore code in Table I.

2. MATERIAL

2.1 Rings and balls. The rings and balls of bearings covered by this specification sheet shall be made of chromium-alloy steel 52100 (UNS 52986).

2.2 Closures. The shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. The number of shields and/or seals required shall be specified in the part number by using the codes listed in Table IV.

2.3 Cage. Cages shall be manufactured from any material unless otherwise specified. The type of cage required shall be specified in the part number by using the codes listed in Table II.

3. METROLOGY

3.1 Radial internal clearance. The tolerance limits for radial internal clearance shall be as specified in AFBMA STD 20 and shall be specified in the part number by using the codes listed in Table 111.

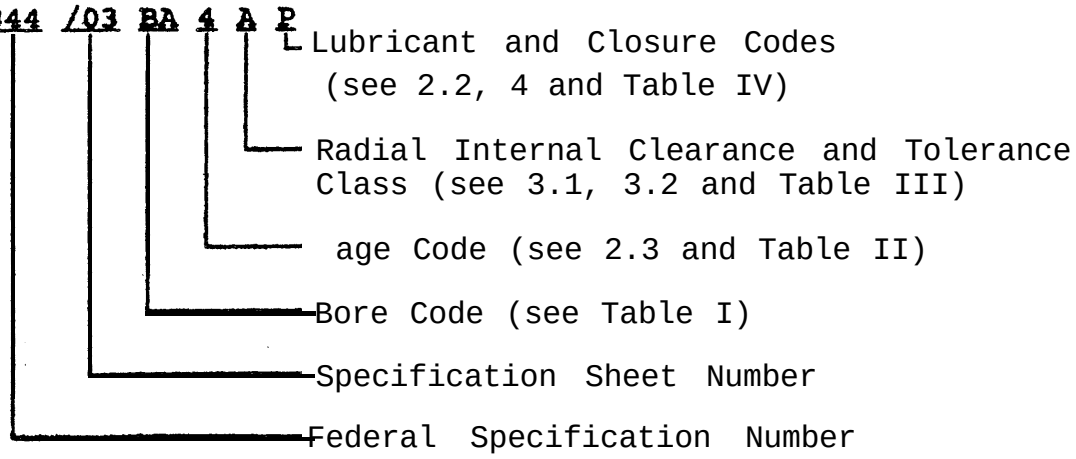
3.2 Tolerance class. The tolerance limits for tolerance class ABEC-1, and ABEC 5, shall be in accordance with ANSI/AFBMA STD 20 and shall be specified in the part number by using the codes listed in Table III.

4. **Lubricant or preservative** compound. The lubricant or preservative compound shall be specified in the part number by using the codes listed in Table IV. Preservative compound shall be applied to all open bearings (no shields or seals and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings, the bearing part number shall include the "C" ‘F” or "L" suffix from Table IV. For double closure or single closure bearings that require lubricant, the bearing part number shall include the preferred suffix from Table IV as required. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled.

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@ 5. Part number. The part number consists of the Federal Specification prefix letters FFB, the basic number of the Federal Specification, the specification sheet number, bore code, and alpha or numeric codes for cage, radial internal clearance and tolerance class, lubricant and closure.

EXAMPLE: FFB2844 /03 BA 4 A P



FFB2844/03BA4AP indicates - Bearing Bore 1.0000 inches; outside diameter 2.0000 inches; width 0.3750 inches; cage type, machined non metallic; radial internal clearance tight; tolerance class ABEC 1; single shield and single seal; lubricant in accordance with DoD-G-24508.

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TABLE I. BOUNDARY DIMENSIONS 1/

Bore Code	0B Bore Diameter (inch)	0B Bore Diameter (MM)	0D Outside Diameter	W Width	Chamfer Min. 2/	Dia. Series 3/	Dynamic Load Rating (lbs)
AA	0.8750	22.2250	1.8750	0.3750	.031	2	2147
AB	0.8750	22.2250	1.8750	0.5000	.031	2	2153
BA	1.0000	25.4000	2.0000	0.3750	.031	2	2147
BB	1.0000	25.4000	2.0000	0.5000	.031	2	2147
BC	1.1250	28.5750	2.1250	0.3750	.031	0	2394
BD	1.1250	28.5750	2.1250	0.5000	.031	0	2394
BE	1.2500	31.7500	2.2500	0.3750	.031	0	2389
BF	1.2500	31.7500	2.2500	0.5000	.031	0	2389
BG	1.3750	34.9250	2.5000	0.4375	.031	0	2794
BH	1.3750	34.9250	2.5000	0.5625	.031	0	2794
BJ	1.5000	38.1000	2.6250	0.4375	.031	0	2936
BK	1.5000	38.1000	2.6250	0.5625	.031	0	2936

1/ All dimensions are in inches unless otherwise specified.
2/ The chamfer on bearings will clear a maximum filet radius equal to the minimum chamfer.
3/ Diameter series is a soft metric conversion of the inch outside diameter and width in comparison with the boundary dimension in ANSI/AFBMA STD 20.

TABLE II. CAGES

CODE	CAGE
1	Pressed Steel or Molded Plastic
2	Machined Brass or Bronze
3	Machined Non Metallic (phenolic)
4	Pressed Brass

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TABLE III. RADIAL INTERNAL CLEARANCE AND TOLERANCE CLASS

Code	Internal Clearance	TOLERANCE
A B C D	SYMBOL 2 (Tight) SYMBOL 0 (Normal) SYMBOL 3 (Loose) SYMBOL 4 (Extra Loose)	ABEC 1
E F G H	SYMBOL 2 (Tight) SYMBOL 0 (Normal) SYMBOL 3 (Loose) SYMBOL 4 (Extra Loose)	ABEC 5

TABLE IV. LUBRICANTS AND CLOSURES

Code	Lubricant	Closure
A B C	DOD-G-24508 MIL-G-81322 Preservative 1/	OPEN (FIG 1)
D E F	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SHIELD (FIG 2)
G H	DOD-G-24508 MIL-G-81322	DOUBLE SHIELD (FIG 3)
J K L	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SEAL (FIG 4)
M N	DOD-G-24508 MIL-G-81322	DOUBLE SEAL (FIG 5)
P R	DOD-G-24508 MIL-G-81322	SINGLE SHIELD AND SEAL (FIG 6)

1/ Preservative shall be in accordance with MIL-P-197.

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